FirstBot: Features and Tuning Guide

A hand-coded Rocket League bot in Java, built on RLBot v5. This guide covers every feature the bot has and every number you can tune.

Status: the code compiles against the real RLBot 3.1.0 library, but it has not been run in a match yet. The aerial, half flip, speed flip and dribble timings are starting values taken from how the mechanics work. Expect to tune them while watching real matches. All tuning constants are at the top of `ExampleBot.java`.

1. How the bot thinks

About 120 times a second, RLBot sends the bot a **game packet** (where the ball and every car are, how fast they're moving, boost amounts) plus a **ball prediction** (where the ball will be over the next few seconds). The bot answers with a **controller state**: throttle, steer, jump, boost, and so on. All of this happens in `getOutput(...)`, in this order:

Step	What happens
1. Kickoff reset	When a kickoff starts, any leftover state from the last play is cleared.
2. Read the situation	The bot estimates when it and the opponent can each reach the ball, checks whether the ball is heading into its own net, predicts where the opponent's shot would go, and checks whether the ball is sitting on its roof. It then picks a **mode** (section 3).
3. Aerial check	If the mode is going for the ball and an airborne ball is reachable, the bot starts an aerial, and the aerial takes over the controls.
4. Pick a target	Each mode chooses a spot to drive to. If boost is low and a pad is close to the path, the bot drives through the pad on the way.
5. Drive	The bot steers toward the target. In dribble mode it balances the ball instead. In shadow and save modes it waits on its spot, facing the ball.
6–7. Actions	Kickoff flips, powerslides, boosting, flipping into the ball, flicks, half flips.
8. Landing	In the air with nothing else to do, the bot levels the car so it lands on its wheels.
9. Maneuvers	Presses the right buttons for any flip that's in progress.

2. Features

Kickoff (speed flip)

The bot boosts toward the ball. About 0.5 s in, it does a **speed flip**: a diagonal front flip that's cancelled by pulling back, so the nose stays level and the bot can keep boosting the whole time. That makes it faster than a plain front flip. Once within 500 units of the ball, it front flips into it. Set `USE_SPEED_FLIP_KICKOFF = false` to go back to the plain front flip.

Driving

The bot steers in proportion to the angle to its target. It works out that angle with `atan2` and wraps it into −π…π so it always turns the short way. It powerslides on very sharp turns and boosts when it's lined up with a far-away target.

Boost management

Rule	Why
Never boost past supersonic (2200 speed)	Top speed is 2300, so boosting there is almost entirely wasted.
Keep a reserve of 20 unless it's urgent	An aerial needs fuel. Saves, challenges and retreats are allowed to dip into the reserve.
Detour through pads near the path	With under 70 boost on a trip over 1500 units, the bot drives through any pad that adds less than 250 units of distance (small pad, +12 boost) or 600 units (big pad, +100 boost).
Go get a big pad when it's safe	The `GET_BOOST` mode (section 3) takes the bot to the nearest big pad on its side of the ball when it's low and the opponent isn't threatening.

Shooting

The bot aims at a point **behind** where it will meet the ball, on the line from the opponent's goal through the ball. Coming in from there, a hit sends the ball toward their goal. The point starts 40% of the distance back (at most 600 units) and closes in as the car approaches, so the car curves onto the line and then drives straight through the ball. When the ball is close, low and straight ahead, the bot front flips into it for power.

Defense (retreating)

If the ball is between the car and its own goal, hitting it would knock it toward the bot's own net. So the bot drives back to the goal post on its side of the ball, which keeps it from driving through the ball. It starts retreating once the ball is 100 units goal-side of it, and stops once it's 300 units goal-side of the ball. Using two different cutoffs stops it flickering between modes.

Aerials

The bot looks through the ball prediction for the first ball above 400 units, 0.6–3 s away. For each one it works out where the car would end up if it jumped and then coasted under gravity:

$$\text{position} = \text{start} + \text{velocity}\cdot t + \tfrac{1}{2}\cdot\text{gravity}\cdot t^2$$

The gap between that point and the hitting spot is what boost has to cover. That takes an average acceleration of $2 \times \text{gap} / t^2$. If that's within 80% of what boost can deliver, there's enough boost, and the ball is roughly in front of the car, the bot jumps. It holds jump for 0.2 s, then double jumps, points its nose along the gap, and boosts while lined up. It recalculates every tick, so it adapts if the ball gets touched.

Air control and landing

`orient(...)` turns the car in the air. It looks at where the aim direction sits relative to the car's nose, roof and right side (using dot products), then sets pitch and yaw from those angles and uses roll to keep the roof

up. It also uses how fast each angle is changing to ease off before overshooting; this is called a PD controller. The same code levels the car out when it's falling, so it lands on its wheels.

Half flip

When the target is more than about 150° behind the car, more than 1500 units away, and the car isn't already driving forward fast, the bot half flips instead of driving a big circle. It backflips, cancels halfway by pushing forward (nose up, roof down), air rolls back upright, and then lets `orient` finish leveling out facing the target.

Dribbling and flicks

When the ball is resting on the roof and the car faces the opponent's half, the bot switches to **dribble** mode. It matches the ball's speed, nudges throttle to keep the ball about 30 units ahead of the car's center, and steers gently toward their goal. When the opponent comes within 900 units, or the car is within 2500 units of their goal, it **flicks**: it jumps, waits a moment so the ball pops up, then does a diagonal front dodge toward the side their goal is on. Diagonal flicks hit harder than straight ones.

3. Opponent awareness (modes)

To estimate when each car reaches the ball, the bot walks the ball prediction and finds the first moment a car's travel time (time to turn plus time to drive) is short enough. Every tick it then checks these rules **top to bottom**, and the first one that matches wins. Watch the console: it prints `Mode: X` every time the mode changes.

Mode	When	What it does
KICKOFF	Kickoff	Speed flip, then flip into the ball
SAVE	Ball predicted to go in our net, or the opponent is about to shoot on target (and we're goal-side)	Hit it if we can get there before it goes in; otherwise stand on the goal line where it will cross
DRIBBLE	Ball on our roof, facing their half	Carry it, then flick
RETREAT	We're on the wrong side of the ball	Drive back to the post
ATTACK	We reach the ball first by at least 0.2 s	Line up behind the ball and shoot
GET_BOOST	They're first, we're below 30 boost, and they aren't a threat	Grab the nearest big pad on our side of the ball
CHALLENGE	They're first, but by less than 0.6 s	Drive straight at the ball to block them
SHADOW	They're first by a lot	Sit 1500 units goal-side of the ball, facing it, and wait for them to commit

To predict the opponent's shot, the bot assumes they'll hit the ball in the direction they're driving into it. It follows that line to our goal line, and if it crosses between the posts (plus a 200-unit margin), the shot counts as on target.

4. Tuning constants

★ marks the constants most likely to need changing first. Change **one at a time**, and watch a few plays after each change.

Opponent awareness

Constant	Default	Raise it →	Lower it →
★ `FIRST_TO_BALL_MARGIN`	0.2 s	attacks less often, plays safer	attacks more, more 50/50s
★ `CHALLENGE_WINDOW`	0.6 s	challenges more	shadows more
★ `SHADOW_DISTANCE`	1500	waits further back	waits closer, more pressure
`HOLD_DISTANCE`	600	starts easing off further from its spot	drives closer to the spot before easing off
★ `LOW_BOOST`	30	goes for big pads more often	rarely goes for big pads
`NO_THREAT_TIME`	1.5 s	stricter about when it's safe to get boost	gets boost even when the opponent is fairly close
★ `SHOT_THREAT_TIME`	0.6 s	reacts to shots earlier, more false alarms	reacts later
`SHOT_WIDTH_MARGIN`	200	treats wider shots as on target	only reacts to shots between the posts
`SAVE_DEPTH`	200	stands further out from the goal line	stands deeper in goal
`SAVE_LOOKAHEAD`	4 s	spots goal-bound balls earlier	only reacts to balls about to go in
★ `TURN_TIME_PER_RADIAN`	0.3 s	assumes turning is slow	assumes turning is quick
`GROUND_HIT_MAX_HEIGHT`	250	counts higher balls as reachable by driving	only low balls count

Boost management

Constant	Default	Raise it →	Lower it →
★ BOOST_RESERVE	20	saves more boost for aerials	spends more on driving
★ PAD_TOP_UP	70	collects pads even when fairly full	only collects pads when nearly empty
PAD_MIN_TRIP	1500	only detours on long trips	detours on short trips too
PAD_MAX_DETOUR_SMALL	250	goes further out of its way for small pads	only takes small pads right on its path
PAD_MAX_DETOUR_BIG	600	goes further out of its way for big pads	only takes big pads near its path
BOOST_MIN_DISTANCE	1500	boosts only for far targets, saves boost	boosts more often

Shooting

Constant	Default	What it does
★ SHOT_OFFSET_RATIO	0.4	How far behind the ball it lines up, as a fraction of the distance. Raise it if shots go off at bad angles.
★ SHOT_OFFSET_MAX	600	The furthest behind the ball it lines up.
FLIP_HIT_DISTANCE	400	Flips into the ball when this close. Too high and it whiffs; too low and it just bumps the ball.
FLIP_HIT_MAX_BALL_HEIGHT	200	Only flips at balls lower than this.
FLIP_HIT_ANGLE	0.25 rad	How well lined up it has to be before flipping (about 14°).

Retreating

Constant	Default	What it does
RETREAT_START	100	How far the ball must be goal-side of the car before the car retreats.
RETREAT_STOP	300	How far goal-side of the ball the car must get before it stops retreating.
RETREAT_POST_X	800	Sideways position of the post it retreats to.
RETREAT_DEPTH	400	How far in front of the goal line it retreats to.

Driving and kickoff

Constant	Default	What it does
STEER_GAIN	2	Higher turns more sharply but can wobble; lower is smoother but slower to turn.
HANDBRAKE_ANGLE	2.0 rad	Powerslides when the target is this far off (about 115°).
BOOST_ANGLE	0.3 rad	Only boosts when lined up within this angle.
USE_SPEED_FLIP_KICKOFF	true	Speed flip kickoff (true) or plain front flip (false).
KICKOFF_FLIP_TIME	0.5 s	When the speed or momentum flip happens.
KICKOFF_FLIP_DISTANCE	500	How close it gets before flipping into the ball.

Flips (times are seconds from the start of the maneuver)

Constant	Default	What it does
FLIP_JUMP_TIME	0.1	How long the first jump is held.
FLIP_DODGE_TIME	0.15	When the second jump (the dodge) happens.
★ SPEED_FLIP_YAW	0.7	Sideways part of the speed-flip dodge (0 = straight, 1 = 45°).
SPEED_FLIP_JUMP_TIME / _DODGE_TIME	0.08 / 0.12	The jump, then the diagonal dodge.
★ SPEED_FLIP_CANCEL_TIME	0.17	When it pulls back to cancel. Too late and the nose dips; too early and it barely flips.
SPEED_FLIP_END_TIME	0.8	When it gives up holding the cancel if it hasn't landed yet.
HALF_FLIP_ANGLE	2.6 rad	How far behind the target has to be (about 150°).
HALF_FLIP_MIN_DISTANCE	1500	Only half flips for targets further away than this.
HALF_FLIP_MAX_SPEED	500	Won't half flip while driving forward faster than this (it powerslides instead).
★ HALF_FLIP_CANCEL_TIME	0.33	When it pushes forward to stop the backflip. It should happen when the car is upside down.
HALF_FLIP_ROLL_TIME	0.3	How long it air rolls before leveling out.

Dribbling and flicks

Constant	Default	What it does
DRIBBLE_MAX_OFFSET	130	The ball counts as "on the car" within this sideways distance.
DRIBBLE_MIN_HEIGHT / _MAX_HEIGHT	90 / 220	...and within this height range above the car's center.
★ DRIBBLE_BALL_AHEAD	30	Where to keep the ball, measured forward from the car's center. If the ball rolls off the front, lower it; if it rolls off the back, raise it.
★ DRIBBLE_SPEED_GAIN	1/200	How strongly it matches the ball's speed.
★ DRIBBLE_POSITION_GAIN	1/40	How strongly it corrects the ball's position.
DRIBBLE_STEER_LEAD	300	How far past the ball, toward their goal, it steers. Higher means gentler turns.
DRIBBLE_STEER_SCALE	0.5	Scales down steering during dribbles so the ball stays on.
★ FLICK_OPPONENT_DISTANCE	900	Flicks when the opponent gets this close.
FLICK_GOAL_DISTANCE	2500	...or when the car is this close to their goal.
★ FLICK_DODGE_TIME	0.25	Dodge timing for flicks. A later dodge lets the ball pop up more first.
FLICK_YAW	0.5	How diagonal the flick is. 0 = straight forward.

Aerials

Constant	Default	What it does
★ AERIAL_ACCEL_MARGIN	0.8	Higher goes for harder aerials (and misses more); lower only goes for easy ones.
★ JUMP_SPEED	800	If it arrives under the ball, raise it; if it arrives over the ball, lower it.
HIT_DISTANCE	110	How far from the ball's center the car aims to be at contact.
AERIAL_MIN_HEIGHT	400	Ignores balls lower than this.
AERIAL_MIN_TIME / _MAX_TIME	0.6 / 3 s	How soon, or how far ahead, an aerial hit can be.
AERIAL_MIN_BOOST	20	Won't jump with less boost than this.
AERIAL_BOOST_ALIGN	0.9	How well lined up the nose must be to boost in the air (0.9 is about 25°).

Air control

Constant	Default	What it does
★ `PITCH_P`, `YAW_P`, `ROLL_P`	3 / 3 / 3	How hard it turns toward the target. Lower these if it **overshoots**.
★ `PITCH_D`, `YAW_D`, `ROLL_D`	1.0 / 1.0 / 0.5	How much it eases off while turning. Raise these if it **wobbles**, lower them if it's **sluggish**.

Game facts (don't change)

`GOAL_Y` 5120, `FIELD_X` 4096, `GOAL_HALF_WIDTH` 893, `GRAVITY` −650, `BOOST_ACCEL` 1060, `BOOST_PER_SECOND` 33.3, `MAX_SPEED` 2300, `MAX_DRIVE_SPEED` 1410, `SUPERSONIC_SPEED` 2200. These values come from Rocket League itself.

5. Troubleshooting

What you see	Try
Flickers between two modes	Widen `CHALLENGE_WINDOW` or `FIRST_TO_BALL_MARGIN`
Too passive, sits back while the opponent dribbles	Lower `SHADOW_DISTANCE`, raise `CHALLENGE_WINDOW`
Gets beaten to the ball in challenges	Raise `TURN_TIME_PER_RADIAN` (it's underestimating how long turns take)
Always out of boost	Raise `BOOST_RESERVE` and `PAD_TOP_UP`; raise `BOOST_MIN_DISTANCE`
Goes for aerials it can't make	Lower `AERIAL_ACCEL_MARGIN`
Aerials arrive under or over the ball	Adjust `JUMP_SPEED`
Wobbles or spins in the air	Raise the `_D` gains, lower the `_P` gains
Speed flip lands on its side or nose	Adjust `SPEED_FLIP_CANCEL_TIME` and `SPEED_FLIP_YAW`
Half flip ends upside down	Adjust `HALF_FLIP_CANCEL_TIME` (cancel when the car is upside down)
Ball falls off during dribbles	Adjust `DRIBBLE_BALL_AHEAD`; lower `DRIBBLE_STEER_SCALE`
Front flips whiff the ball	Lower `FLIP_HIT_DISTANCE`

To practice one situation over and over, such as an aerial or a save, use RLBot's state setting or a training pack.

6. Not built yet

Mechanic	Notes
Flip resets	Getting all four wheels onto the ball mid-air resets the car's flip, so it can dodge again. This needs very precise air control: matching the car's roof direction to the ball's surface at exactly the right moment. It's worth trying once aerials hit reliably.
Jump hits	Balls between about 250 and 400 units high: too high to drive into, too low to be worth an aerial. These need a timed single or double jump.
Getting the ball onto the roof	Right now the bot only dribbles when the ball happens to land on the roof. A "catch" or "pop-up" routine would let it start dribbles on purpose.
Wavedash	A dodge as the wheels touch the ground after a jump, for a quick burst of speed while landing.
Ceiling shots, wall play	The bot treats the field as flat; walls and ceiling aren't used on purpose.
Demolitions	Hitting the opponent at supersonic speed to knock them out of the play briefly.
Reinforcement learning	Stretch goal: train a bot with RLGym and compare it to this hand-coded bot.

7. Sources

Speed flip mechanics: thespike.gg "How to speed flip" (https://www.thespike.gg/rocket-league/beginner-guides/how-to-speed-flip); upcomer.com "How to perform the Speed Flip in Rocket League" (https://upcomer.com/how-to-perform-the-speed-flip-in-rocket-league/).

Half flip mechanics: mp1st.com "Rocket League Half Flip – How to Master" (https://mp1st.com/news/rocket-league-half-flip-how-to-master).

Game values (supersonic 2200, max speed 2300, boost use 33.3/s, boost acceleration ~992, gravity 650): RLGym game values cheatsheet (https://rlgym.org/Cheatsheets/game_values).

RLBot framework: https://rlbot.org and https://github.com/RLBot/java-interface.